

The Crypto Fear Channel: Asymmetric, Tail-Concentrated, and Regime-Dependent Volatility Spillover from Bitcoin to Equity Markets

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Abstract

Using daily returns on SPY, BTC-USD, and VIX from 2015-02 to 2026-04 ($N = 2,812$), we document three stylized facts about the crypto-to-equity volatility spillover. First, the spillover is *asymmetric*: Bitcoin downside volatility Granger-causes VIX at lags 1–5, whereas Bitcoin upside volatility does not (symmetric Granger tests covering lags 1–10 corroborate the broader directional result). Second, the spillover is *tail-concentrated with sign reversal*: the quantile-regression coefficient of VIX on BTC realized variance is significantly *negative* at lower quantiles (-2.86 at $\tau = 0.05$; -2.34 at $\tau = 0.25$), indicating bull-market decoupling, then turns positive at the median ($+2.61$) and amplifies to $+22.31$ at the 95th percentile, yielding an $8.5\times$ upper-tail amplification relative to the median. Third, the spillover is *regime-dependent*: in a five-subperiod breakdown, Granger causality is statistically significant only during 2020 ($F = 11.05$, $p < 0.001$) and non-significant in 2015–2017, 2018–2019, 2021–2022, and 2023–2026. In the Diebold-Yilmaz framework, Bitcoin is a *net receiver*, rehabilitating the narrative from “crypto as a fear originator” to “crypto as a fear amplifier” conditional on pre-existing equity stress. Out-of-sample forecasts, however, show that adding BTC realized volatility does not statistically improve an AR model of VIX (Diebold-Mariano $t = -0.98$, $p = 0.33$), highlighting that Granger causality is a necessary but not sufficient condition for forecastability. Taken together, our findings reframe the crypto-equity spillover as a *crisis-time amplifier*: informative for ex-post risk attribution and margin design, but inadequate as an ex-ante predictor of equity fear.

Keywords: volatility spillover, Bitcoin, VIX, asymmetric causality, quantile regression, Diebold-Yilmaz, crypto-equity linkage

JEL: G11, G15, G17, C22

1 Introduction

Since the institutionalization of Bitcoin trading and the 2024 approval of U.S. spot BTC ETFs, the question of whether cryptocurrency markets transmit fear to traditional equity markets has acquired first-order practical importance for portfolio risk management, prudential supervision, and margin design. Yet the empirical literature has yielded an uneasy consensus. On one hand, correlation-based studies report rising BTC-equity co-movement in stress periods, suggesting that cryptocurrency is no longer a standalone asset class.¹ On the other hand, return-predictability tests often fail to find that Bitcoin-based signals improve forecasts of equity volatility at practical horizons. This tension motivates a central question: is the crypto-equity fear channel *real but nonlinear*, *real but regime-dependent*, or an artifact of tail-period correlation without predictive content?

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¹See, e.g., Bouri et al. (2020), Corbet et al. (2018), and Matkovskyy and Jalan (2019) for the early post-2017 literature.

We resolve this tension by decomposing the spillover along three dimensions that the prior literature has typically examined in isolation: directional asymmetry, tail dependence, and regime stability. Each of these dimensions yields a stylized fact that materially changes how one should interpret correlation-based evidence.

Asymmetry. Using the asymmetric Granger-causality framework of Hatemi-J (2012), we separate Bitcoin positive and negative realized volatility and test whether each Granger-causes VIX at lags 1 through 5. We find that the negative branch carries the entire spillover: Bitcoin downside volatility Granger-causes VIX at every lag tested (1–5), while the upside branch is uniformly non-significant across the same lag window. This is not merely a statistical artifact of a leverage effect — the asymmetry is corroborated by standard symmetric Granger tests over the broader 1 through 10 lag range, where BTC-to-VIX causality is significant at lags 2–10. Our interpretation is behavioral: retail-heavy crypto markets transmit *fear*, not *euphoria*, into the institutional equity complex.

Tail concentration with sign reversal. Quantile regressions of VIX on BTC realized variance reveal a richer structure than a simple tail amplification: the slope *reverses sign* across the VIX distribution. At the lower quantiles the coefficient is significantly *negative* (-2.86 at $\tau = 0.05$; -2.34 at $\tau = 0.25$), consistent with a bull-market coexistence regime in which rising Bitcoin volatility coincides with *compressing* equity fear. The slope turns positive at the median ($+2.61$ at $\tau = 0.5$) and amplifies sharply in the upper tail ($+22.31$ at $\tau = 0.95$), an approximately $8.5\times$ amplification of the upper tail relative to the median ($\tau = 0.95/\tau = 0.5 \approx 8.54$). The spillover is therefore not a uniform channel, nor even a simple “flat-then-spikes” tail phenomenon, but a regime with a *directional switch* at the median of VIX: BTC volatility dampens equity fear in calm markets and amplifies it in panic markets. Conventional OLS or DCC-GARCH estimates — which collapse this sign-reversing heterogeneity into a single slope — systematically mischaracterize both the bull-market decoupling and the crisis-time amplification of crypto volatility on equity fear.

Regime dependence. A five-subperiod breakdown reveals that Granger significance is concentrated entirely in 2020 ($F = 11.05$, $p < 10^{-6}$). Pre-mania (2015–2017), crypto-winter (2018–2019), bull-bear (2021–2022), and recovery-plus-ETF (2023–2026) all fail to reject non-causality. Combined with the finding from Diebold and Yilmaz (2012) spillover-index analysis that Bitcoin is a *net receiver* rather than net sender, this suggests that the spillover is not a permanent feature of cross-market architecture but a *conditional amplification mechanism* that activates only during pre-existing equity stress.

The forecastability gap. A natural next question is whether these three stylized facts translate into out-of-sample predictive power. They do not. Comparing an $AR(p)$ specification of VIX against one augmented with BTC realized volatility over an OOS window of 1,826 days (2019–2026), the Diebold-Mariano test returns $t = -0.98$, $p = 0.33$ — statistically indistinguishable from zero, and below the Harvey et al. (2016) $|t| > 3$ threshold for newly proposed factors. This result echoes a broader methodological point: *Granger causality is a necessary but not sufficient condition for forecastability*. An informative signal that appears only in tail events and only during rare regimes may be too sparse to outperform a well-specified autoregressive benchmark on average over a long OOS sample.

We make three contributions to the literature. First, by combining asymmetric Granger causality, quantile regression for tail dependence, Diebold-Yilmaz spillover direction, and an honest out-of-sample forecasting test in a single framework, we show that each dimension tells a distinct story that prior single-technique studies have missed. Second, the regime-dependent finding — that COVID-2020 is a structural watershed — implies that volatility spillover from

crypto is a *crisis-time amplifier* rather than a steady-state channel, with direct implications for margin and capital requirements during regime transitions. Third, the predictive-power null, reported transparently rather than hidden, contributes to a growing methodological awareness in the volatility forecasting literature that in-sample significance and out-of-sample usefulness can diverge substantially (Harvey et al., 2016; Conrad and Kleen, 2020).

The remainder of the paper is organized as follows. Section 2 surveys the crypto-equity spillover literature and positions our contribution. Section 3 describes the data and preliminary diagnostics. Section 4 details the five methodological building blocks. Sections 5 and 6 present the main results and robustness checks. Section 7 reports the out-of-sample forecasting evaluation. Section 8 discusses mechanisms and policy implications. Section 9 concludes.

2 Literature Review

We organize the related literature along three threads: (i) the empirical literature on crypto-equity volatility spillover and the safe-haven debate, (ii) the methodological literature underlying our four building blocks (asymmetric Granger, quantile regression, Diebold-Yilmaz, Diebold-Mariano), and (iii) the recent calls for honest out-of-sample evaluation in volatility forecasting. Each thread has progressed independently; our contribution is to combine them within a single framework that decomposes the spillover into asymmetry, tail concentration, regime conditionality, and predictive power.

2.1 Crypto-equity volatility spillover and the safe-haven debate

The empirical literature on Bitcoin’s role in equity portfolios has evolved through three phases. The first phase, exemplified by Corbet et al. (2018) and Matkovskyy and Jalan (2019), documented that cryptocurrency price movements respond meaningfully to macro-financial news and that contagion-style co-movement with equity markets emerges in stress periods, contradicting the early “digital gold” framing. The second phase, motivated by COVID-19 and reviewed in Bouri et al. (2020), Conlon and McGee (2020), Shahzad et al. (2019), and Klein et al. (2018), sharpened the safe-haven question with mixed verdicts: wavelet- and DCC-based correlation analysis showed that Bitcoin’s risk profile is closer to a high-volatility risky asset than to gold, and that its safe-haven properties are at best regime-dependent. The third phase, post-2020, has documented increasing institutional integration: Yarovaya et al. (2022) report that information transmission between cryptocurrency and traditional asset markets intensified during the pandemic, and Akyildirim et al. (2020) link implied-volatility shocks to cryptocurrency returns.

Three gaps motivate our analysis. First, most prior studies test *symmetric* causality or *symmetric* correlation, conflating the upside and downside volatility channels that we will show carry economically distinct information. Second, the tail dependence of the spillover is typically inferred indirectly from rolling DCC or wavelet coherence, rather than directly estimated through quantile regression. Third, sample windows often end at 2020 or include 2020 as a single observation, leaving the post-COVID regime-stability question unaddressed — a gap that matters precisely because COVID-2020 will turn out to be the structural watershed of the spillover. Our analysis closes these gaps by replacing each symmetric or correlation-based summary with a direct, regime-aware, asymmetric counterpart, while extending the sample through April 2026.

2.2 Methodological building blocks

Our four building blocks each have a settled methodological literature. Asymmetric Granger causality, the technique behind our headline asymmetry result, was formalized by Hatemi-J (2012), who decompose each series into positive and negative cumulative innovations and test their respective Granger predictability. Earlier nonparametric extensions of Granger causality (Diks and Panchenko, 2006) and asymmetric volatility-spillover tests (Hong, 2001) provide

complementary diagnostic tools that we use as robustness checks. Each of these methods was developed to address the specific concern that linear Granger tests can mask sign-dependent transmission — precisely the asymmetry the present paper documents in the BTC-to-VIX channel.

Quantile regression, our tool for tail dependence, originates with Koenker and Bassett (1978) and has subsequently been extended into systemic-risk applications. Most relevant to our setting is Adrian and Brunnermeier (2016)’s CoVaR framework, in which the conditional value-at-risk of one institution given another’s stress is estimated through quantile regression on returns. We adopt the quantile-regression framework rather than CoVaR *per se* because our object of interest is the conditional response of equity *volatility* (VIX), not the conditional value-at-risk of equity returns; but the underlying statistical engine is the same.

The Diebold-Yilmaz spillover index is now a canonical tool for measuring directional volatility transmission across asset classes. Its development spans Diebold and Yilmaz (2009), Diebold and Yilmaz (2012), and Diebold and Yilmaz (2014), and a recent IMF policy note (Iyer, 2022) applies the framework specifically to cryptocurrency-equity spillovers. We follow the 2012 specification (variance decomposition of a generalized VAR) and report rolling 252-day spillover indices to assess directional connectedness over time.

Finally, our forecasting evaluation rests on the Diebold and Mariano (1995) test of equal predictive accuracy and the Harvey et al. (2016) multiple-testing threshold of $|t| > 3$ for newly proposed factors. The latter has become a methodological norm for newly proposed predictors in asset pricing, and we adopt it in the volatility-forecasting setting to avoid the inflated false-positive rate associated with conventional $|t| > 1.96$ inference under multiple testing.

2.3 Honest out-of-sample evaluation in volatility forecasting

A recurring tension in the volatility-forecasting literature is the gap between in-sample statistical significance and out-of-sample economic value. Conrad and Kleen (2020)’s GARCH-MIDAS application demonstrates that incorporating macro-economic information through long-run components can deliver in-sample fit improvements that fail to translate into forecast accuracy gains. The cumulative evidence reviewed there motivates an evaluation discipline in which a new predictor must clear both (i) in-sample significance under the Harvey threshold and (ii) out-of-sample improvement under Diebold-Mariano, before being promoted to a structural claim. The present paper adopts this discipline and reports a transparent out-of-sample null — BTC realized volatility fails to improve AR-VIX forecasts at $t = -0.98$ — alongside its in-sample asymmetry, tail-amplification, and regime-stability findings. The combination is, to our knowledge, the first such honest joint reporting in the crypto-equity spillover literature.

3 Data and Preliminaries

3.1 Sample and variables

Our sample consists of daily data on three series: the SPDR S&P 500 ETF (SPY), Bitcoin in U.S. dollar terms (BTC-USD), and the CBOE Volatility Index (VIX). All series are obtained from Yahoo Finance via the `yfinance` package and pinned to a fixed snapshot CSV (`paper/crypto-fear-channel/data/spy_btc_usd_vix_2015-2026.csv`, 4,116 rows including BTC’s seven-day calendar) generated on 2026-04-27 with `auto_adjust=False`, retaining both raw close and adjusted close columns to insulate replication from retroactive dividend reconciliations. The full sample spans 2015-02-02 through 2026-04-08, yielding $N = 2,812$ trading-day observations after aligning BTC to the SPY/VIX trading calendar (we drop weekend BTC observations because VIX is undefined). The out-of-sample window used in §4.5 begins 2019-01-01 and runs through 2026-04-08, providing 1,826 OOS observations — spanning the COVID-19

shock (2020), the late-cycle bull market (2021), the 2022 bear, and the post-ETF recovery (2023–2026).

3.2 Constructed variables

We work with three constructed series. Daily log returns are computed from adjusted close: $r_t^{\text{spy}} = \log(P_t^{\text{spy}}/P_{t-1}^{\text{spy}})$ and $r_t^{\text{btc}} = \log(P_t^{\text{btc}}/P_{t-1}^{\text{btc}})$. Realized volatility is constructed as a 22-day rolling standard deviation of log returns, annualized by $\sqrt{252}$: $\text{RV}_t^{(20)} = \text{std}_{[t-21,t]}(r) \cdot \sqrt{252}$, computed separately for SPY and BTC. The VIX is used in level (it is already an annualized measure, %). For all forward-looking specifications we lag explanatory variables by one day (`signal.shift(1)` convention) to enforce causal predictability and rule out look-ahead.

For the asymmetric Granger tests of §4.2 we further decompose BTC realized volatility into two non-negative branches following Hatemi-J (2012): $\text{RV}_t^{\text{btc},+} = \max(\Delta \text{RV}_t^{\text{btc}}, 0)$ and $\text{RV}_t^{\text{btc},-} = -\min(\Delta \text{RV}_t^{\text{btc}}, 0)$. The cumulative positive and negative innovations are then tested separately for Granger predictability of VIX.

3.3 Descriptive statistics and stationarity

Table 1 reports descriptive statistics for the four primary series. BTC daily returns exhibit mean 0.23%, daily standard deviation 3.76%, and excess kurtosis 7.58 — materially fatter-tailed than SPY (14.15 excess kurtosis is concentrated in a handful of crisis days), reflecting the well-documented heavy-tailed nature of cryptocurrency returns. The VIX averages 18.4 with maximum 82.7 (March 2020) and minimum 9.1, spanning the calm 2017–2019 vol regime, the COVID shock, the 2022 inflation regime, and the 2024–2026 normalization.

Table 1: Descriptive statistics for daily series, 2015-02-02 to 2026-04-08 ($N = 2,812$).

Series	Mean	Std. dev.	Skewness	Excess kurtosis
r^{btc} (%)	0.229	3.764	−0.093	7.579
r^{spy} (%)	0.056	1.117	−0.307	14.150
VIX	18.382	7.110	—	—
$\text{RV}_{\text{btc}}^{(20)}$	0.542	0.257	—	—

VIX min/max are 9.14 / 82.69. $\text{RV}_{\text{btc}}^{(20)}$ min/max are 0.098 / 1.701 (annualized). Source: `experiments/k1025/k1025_results.json .descriptive_statistics`.

Augmented Dickey-Fuller (ADF) stationarity tests reject the unit-root null for all three modeling series at conventional levels: $\text{RV}_{\text{btc}}^{(20)}$ ($t = -4.90$, $p < 0.001$, lags 20), VIX ($t = -5.71$, $p < 10^{-6}$, lags 9), and $\text{RV}_{\text{spy}}^{(20)}$ ($t = -4.46$, $p < 0.001$, lags 20). Stationarity is a precondition for the Granger and Diebold-Yilmaz analyses below, both of which assume covariance stationarity. We do not differentiate further; the analyses use the level series.

4 Methodology

We employ four methodological building blocks. The first three (§4.1–4.4) characterize *in-sample* cross-asset structure along three dimensions: directional asymmetry, tail dependence, and regime stability. The fourth (§4.5) tests *out-of-sample* predictive content.

4.1 Symmetric Granger causality (baseline)

As a baseline we test whether BTC realized volatility Granger-causes VIX in a standard linear specification with lags $\ell = 1, \dots, 10$. For each lag ℓ we estimate the unrestricted regression

$$\text{VIX}_t = \alpha_0 + \sum_{j=1}^{\ell} \alpha_j \text{VIX}_{t-j} + \sum_{j=1}^{\ell} \beta_j \text{RV}_{\text{btc}, t-j}^{(20)} + \varepsilon_t \quad (1)$$

and the restricted regression that drops the β_j terms; the joint null $H_0 : \beta_1 = \dots = \beta_{\ell} = 0$ is tested with a standard F -test under HAC standard errors. We tabulate F -statistics and p -values across lags 1 through 10 and treat the symmetric Granger result as the linear-causality benchmark against which the asymmetric and tail-conditional specifications below are compared.

4.2 Asymmetric Granger causality (Hatemi-J)

Following Hatemi-J (2012), we decompose BTC realized-volatility innovations into positive and negative cumulative components and test their separate Granger predictability of VIX:

$$\text{VIX}_t = \alpha_0 + \sum_{j=1}^{\ell} \alpha_j \text{VIX}_{t-j} + \sum_{j=1}^{\ell} \beta_j^+ \text{RV}_{t-j}^{\text{btc}, +} + \sum_{j=1}^{\ell} \beta_j^- \text{RV}_{t-j}^{\text{btc}, -} + \varepsilon_t. \quad (2)$$

We test two separate null hypotheses, $H_0^+ : \beta_1^+ = \dots = \beta_{\ell}^+ = 0$ (BTC upside does not Granger-cause VIX) and $H_0^- : \beta_1^- = \dots = \beta_{\ell}^- = 0$ (BTC downside does not Granger-cause VIX), each with a separate F -test on the asymmetric residual decomposition. The Hatemi-J framework is appropriate when the suspected transmission channel is sign-dependent — our prior, motivated by retail-driven crypto trading and the asymmetric leverage effect in equity volatility, is precisely such asymmetry. We test lags $\ell \in \{1, 2, 3, 4, 5\}$.²

4.3 Quantile regression for tail dependence

Linear regression measures the conditional mean response of VIX to BTC RV; conditional on the documented heavy tails of both series and the sign-reversing structure we will document, the conditional mean is uninformative about regime-specific transmission. We follow Koenker and Bassett (1978) and estimate quantile regressions of VIX on BTC realized variance at four representative quantiles $\tau \in \{0.05, 0.25, 0.50, 0.95\}$:

$$Q_{\tau}(\text{VIX}_t \mid \text{RV}_{\text{btc}, t-1}^{(20)}) = \alpha_{\tau} + \beta_{\tau} \cdot \text{RV}_{\text{btc}, t-1}^{(20)}, \quad (3)$$

estimated via the standard Koenker-Bassett check-loss minimization. Standard errors and confidence intervals are obtained via 1,000-replication bootstrap of the regression sample. The tail-dependence ratio reported in our headline result is the upper-tail amplification $\beta_{0.95}/\beta_{0.50}$, which exceeds 8 in our sample. We also report intermediate quantiles in robustness checks. The conceptual link to Adrian and Brunnermeier (2016)'s CoVaR framework is direct: both rely on quantile regression to recover state-dependent cross-asset transmission, with the present paper applying the technique to volatility (VIX) rather than tail risk.

4.4 Diebold-Yilmaz spillover index

To measure directional volatility transmission, we follow the Diebold and Yilmaz (2012) variance-decomposition specification. We fit a generalized vector autoregression of order $p = 4$ to the

²The lag window 1–5 is chosen to match the timescale at which retail-investor sentiment plausibly transmits across asset markets; the symmetric Granger test in §4.1 extends to $\ell = 10$ to confirm robustness across longer horizons.

3-variable system $\{VIX, RV_{\text{spy}}^{(20)}, RV_{\text{btc}}^{(20)}\}$ and compute the H -step-ahead generalized forecast-error variance decomposition (FEVD) at $H = 10$ days. The directional spillover from variable i to variable j is the share of variable j 's H -step forecast-error variance attributable to shocks originating in variable i . The total spillover index, the “to” index, the “from” index, and the net spillover (to minus from) are computed in the standard generalized framework. To track time variation, we compute the same indices on a rolling 252-trading-day window, allowing direct visualization of how the BTC–VIX–SPY spillover structure evolves across the 2015–2026 sample.

4.5 Out-of-sample forecasting framework

We test whether the in-sample asymmetry, tail-amplification, and regime-stability findings translate into out-of-sample predictive power for VIX. The base forecast is an autoregressive $AR(p)$ model with lag length selected by AIC on the in-sample window (2015-02 to 2018-12); the augmented forecast adds lagged BTC realized volatility:

$$VIX_t = \alpha_0 + \sum_{j=1}^p \alpha_j VIX_{t-j} + \gamma \cdot RV_{\text{btc}, t-1}^{(20)} + u_t. \quad (4)$$

Both models are re-estimated on a rolling basis, and one-day-ahead point forecasts are generated for each day in the OOS window 2019-01-01 to 2026-04-08 (1,826 observations). Forecast errors are evaluated under squared-error loss, and the difference in loss between the two specifications is tested with the Diebold and Mariano (1995) test of equal predictive accuracy under the Harvey et al. (2016) threshold $|t| > 3$ for newly proposed predictors — a tighter discipline than the conventional $|t| > 1.96$ benchmark, motivated by the multiple-testing concerns reviewed in Conrad and Kleen (2020). Failure to reject the null at this threshold is interpreted as a transparent signal that, despite in-sample causal structure, BTC realized volatility does not improve VIX point forecasts on average over the long OOS sample.

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